

Manthan Sunil Shende

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PROFESSIONAL SUMMARY

Aspiring IoT Engineer and Software Developer currently pursuing a B.Tech in Industrial IOT. Passionate about embedded systems, sensor integration, and building intelligent technology solutions for real-world applications. Alongside a strong interest in web development, and IoT-based technologies. Eager to apply technical knowledge, contribute to innovative projects, and continuously grow as a technology professional.

EDUCATION

B.Tech. in Industrial Internet of Things

St. Vincent Pallotti College of Engineering and Technology, Nagpur

Pursuing: 2028

CGPA: 9.06 / 10

SKILLS SUMMARY

- **IoT & Embedded Systems:** ESP32, Arduino, Sensor Integration, 8085/8086 Microprocessor, 8051 Microcontroller, 8085/8051 Assembly Language Programming (ALP)
- **Languages:** C, Python, Embedded C, R Programming
- **Web Technologies:** HTML5, CSS3, JavaScript, React
- **Developer Tools:** VS Code, GitHub, Git, Antigravity, Arduino IDE, HiveMQ Cloud

INTERNSHIP EXPERIENCE

EMBEDDED SYSTEM INTERN — INTERNPE

Jun 2026 – Jul 2026

- Completed a 4-week internship focused on Embedded Systems fundamentals and practical applications.
- Worked with microcontrollers, embedded programming concepts, and hardware-software interfacing.
- Gained hands-on exposure to embedded system design, sensor integration, and real-time system development.
- Strengthened knowledge of Industrial IoT, microcontroller architectures, and embedded development workflows.

PROJECTS

Smart Radar System — Arduino Uno, Ultrasonic Sensor, Servo Motor, Embedded C

2025

- Developed a smart radar system using Arduino Uno, Ultrasonic sensor, and servo motor for real-time object detection and distance measurement.
- Implemented ultrasonic sensing techniques to calculate object distance by measuring the time-of-flight of sound waves.
- Programmed servo motor-based scanning from 0° to 180° to provide wide-area object detection similar to radar systems.
- Processed and transmitted real-time sensor data through serial communication for dynamic monitoring and visualization.
- Applied concepts of embedded systems, sensor interfacing, microcontroller programming, and real-time data acquisition.

Voice-Controlled Robot Car — ESP32, Bluetooth, L298N Motor Driver, Arduino IDE

2025

- Developed a voice-controlled robotic car using ESP32 and Bluetooth communication for hands-free navigation and wireless control.
- Integrated an Android voice-control application to convert speech commands into movement instructions, enabling forward, backward, left, right, and stop operations.
- Implemented motor control using the L298N motor driver and DC gear motors, ensuring accurate directional movement and speed regulation.
- Simulated and tested the system using Tinkercad before hardware implementation, improving reliability and reducing development errors.
- Applied concepts of embedded systems, Bluetooth communication, robotics, motor interfacing, and real-time control systems.

Posture Guard AI Web — HTML, CSS, JavaScript

2026

- Developed a web-based posture monitoring application to promote healthy sitting habits and improve ergonomic awareness.
- Designed a responsive and user-friendly interface using HTML, CSS, and JavaScript for real-time posture tracking.
- Implemented posture analysis and alert mechanisms to notify users when poor posture is detected.
- Focused on accessibility, usability, and real-time feedback to enhance the overall user experience.

CERTIFICATIONS & ACHIEVEMENTS

- **Participant, Central India Hackathon (CIH) 2.0 (2025)** – Collaborated in a team-based innovation challenge focused on solving real-world technical problems.
- **Participant, HVAC&R-Based Hackathon (2025)** – Contributed to the development of technology-driven solutions in the Heating, Ventilation, Air-Conditioning, and Refrigeration domain.
- **API Development Workshop** – Gained hands-on experience with REST APIs, Node.js, Express.js, Postman, JSON, HTTP methods, and backend development concepts. (WebGurukul IT Training & Placement, 2026)